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Synergistic effects of rice husk ash and neutral silicate activator on rheology and strength of ambient-cured self-compacting geopolymer concrete

Thatikonda Naresh^{1*}, Mainak Mallik², S. Venkateswara Rao³, Md. Sameer⁴, Rajesh Kumar Dora⁵, V. V. Praveen Kumar⁶ and N. Satya Shiva Prasad⁷

*Correspondence:

Thatikonda Naresh

naresh.phd20@nitap.ac.in

¹Department of Civil Engineering
ACE Engineering College,
Hyderabad, India

²Department of Civil Engineering,
National Institute of Technology,
Jote, Arunachal Pradesh, India

³Department of Civil Engineering,
National Institute of Technology,
Warangal, India

⁴Department of Mechanical
Engineering, Kakatiya Institute of
Technology and Science Warangal,
Warangal, India

⁵Department of Mechanical
Engineering, Samata
Chandrasekhar Institute of
Technology and Management
SCITM, Similiguda, Koraput, Odisha,
India

⁶Department of Civil Engineering,
Geethanjali College of Engineering
and Technology, Hyderabad, India

⁷Department of Civil Engineering,
Kakatiya University, Warangal, India

Abstract

This study investigated the development of ambient-cured self-compacting geopolymer concrete ((SCGC)) using a neutral-grade sodium silicate solution as the sole activator. The binder system comprised ground granulated blast furnace slag (GGBFS), fly ash (FA), and micro silica (MS), with partial substitution of MS by rice husk ash (RHA) at 5–15% to improve sustainability and pozzolanic reactivity. Mixes were evaluated at two binder contents (500 and 600 kg/m³) and two solution-to-binder (S/B) ratios (0.6 and 0.5), with assessments of fresh properties, mechanical performance, and microstructural characteristics. All formulations demonstrated satisfactory workability and cohesive rheological behavior. The early age strength development was significant, with selected RHA-based mixes achieving over 50% of their 28-day compressive strength within 3 days and exceeding 85% by 7 days. The optimal mix, incorporating 600 kg/m³ binder and 15% RHA at 0.5 S/B, exhibited a 26% strength gain over the control mix, with corresponding improvements in tensile and flexural strengths. The SEM analysis revealed dense gel matrices with interpenetrating N–A–S–H and C–A–S–H phases, reduced porosity, and improved gel continuity. The results demonstrate that ambient-cured SCGC based on a hybrid industrial-waste binder constitutes a promising, lower-clinker geopolymer binder system with enhanced fresh, mechanical, and microstructural performance, while long-term durability and structural behavior require further investigation.

Keywords Self-Compacting geopolymer concrete, Single activator solution, Fresh property, Micro silica (MS), Compressive strength

1 Introduction

Infrastructure development in emerging countries has accelerated across multiple sectors to fulfill the growing demand of industries, particularly in metropolitan regions [1]. Cement concrete is used more frequently because of the increase in construction [2]. However, because of the carbon dioxide generated during cement manufacturing,



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higher cement use contributes to elevated atmospheric CO₂ emissions and air pollution [3]. Concurrently, cement production depletes natural rock deposits and requires significant energy, whereas calcination operations require temperatures as high as 1500 °C [4]. An interesting strategy is to use industrial and agricultural waste materials that are abundant in India and around the world, such as Fly Ash (FA), met kaolin, Ground Granulated Blast Furnace Slag (GGBFS), Rice Husk Ash (RHA), and silica fume (SF) [5]. These minerals have pozzolanic qualities and have long been employed as additional elements in cement to improve its flexibility, potency, and longevity [6]. Researchers have explored the viability of replacing Portland cement with FA in Self-compacting concrete (SCC) and discovered that this substitution improves the features of concrete that are both fresh and hardened [7].

Furthermore, these waste materials, combined with naturally occurring minerals such as clay and clay kaolinite, can be used to produce geopolymer concrete (GPC) [8]. GPC, a sustainable alternative to conventional Portland cement-based concrete, generates inorganic binders via alkaline solution activation [9]. Because geopolymer-based concrete does not use cement and requires less calcination energy than traditional cement manufacturing, it significantly reduces CO₂ emissions [10]. Geopolymer cement uses silicon and an aluminum-rich resource, frequently derived from waste goods, and converts them into environmentally friendly building materials, thereby reducing industrial waste and emissions [11]. Thus, geopolymer technology contributes to environmental sustainability goals by lowering the CO₂ emissions from cement production [12]. However, curing techniques affect the mechanical qualities of GPC, and high-temperature curing processes require a large amount of energy [13].

Extended curing times can disturb granular structures, reducing the overall strength, whereas early stage exposure to high temperatures can result in larger pores and poorer material characteristics [14]. To solve these issues, researchers have investigated different mixtures for substituting Fly Ash for GGBFS, removing the need for thermal curing, and increasing the compressive strength [15]. Self-compacting concrete (SCC) eliminates the need for external compaction technologies, providing benefits such as increased compaction, decreased noise pollution, and lower construction costs [16]. Self-compacting geopolymer concrete (SCGC) incorporates the advantages of geopolymer technology with S.C.C., reducing the need for compaction and eliminating the use of Portland cement [17]. These breakthroughs have paved the way for more sustainable concrete manufacturing, with research focusing on materials such as Rice Husk Ash (RHA), an abundant agro-industrial waste product in India, is a potential substitute for improving the quality of GPC [18]. Ambient temperature curing of FA-based geopolymer mortars was achieved by the addition of calcium hydroxide and oxide [19]. This was because the accumulation of calcium compounds enhanced their mechanical characteristics. To increase the early strength of fly ash by introducing CaO-containing materials, such as GGBFS or nanomaterials, as partial replacements for FA [20].

Several studies have examined how precursor chemistry and curing regimes affect alkali-activated geopolymer concretes. Nis and Altindal [21] showed that FA-based binders are highly sensitive to the curing temperature. Zhang et al. [22] demonstrated that waste-ceramic slag systems activated with Na₂SO₃/NaOH improved the mechanical and durability properties. Hwalla et al. [23] reported that geopolymer screeds exhibit higher strength and impact resistance than cement-based screeds. Zhang et al. [24] observed

that slag–FA binders containing ceramic powders showed improved thermal stability at elevated temperatures. Sherwani et al. [25] confirmed that increasing the GGBFS content enhanced the fresh and hardened performance of FA-based SCGC. Heweidak et al. [26] supported the suitability of SCGC for composite slab systems owing to its improved bond behavior and reduced carbon footprint. Furthermore, Sherwani et al. [4] found that slag-rich SCGC provide superior mechanical performance and durability, with the curing temperature playing a critical role.

Recent advancements in sustainable alkali-activated and SCM-based systems have provided critical scientific insights that directly complement the present study. Rashidi et al. [27] conducted a comprehensive multi-scale assessment of Ferro-silicomanganese/GGBFS-based alkali-activated composites and demonstrated that the evolution of gel morphology from fibrous high-tortuosity networks to fully cross-linked 3D C–(N)–A–S–H structures plays a decisive role in governing the mechanical resistance and transport behavior. Their findings underscore the necessity of understanding gel topology when developing high-performance alkali-activated binders. Bameri et al. [28] further highlighted the synergistic benefits of multi-component SCM systems, showing that hybrid blends of silica fume, waste glass powder, and GGBFS significantly enhance strength, pore refinement, and chloride resistance owing to improved packing density and intensified pozzolanic activity. In a related contribution, Bameri et al. [29] demonstrated that optimizing waste glass powder content not only improves long-term mechanical performance but also substantially reduces autoclave expansion, embodied carbon, and energy consumption, thereby reinforcing its role as a high-value sustainable SCM. Rashidi et al. [30] also reported that the combined use of pumice powder and silica fume in fiber-reinforced composites produces dense C–S–H/C–A–S–H matrices, leading to remarkable enhancements in the compressive strength, stiffness, and bond performance, which are clear indicators of the microstructural advantages of dual-SCM systems. Additionally, Kanagaraj et al. [31] proposed a structured mix-design methodology for superplasticizer-free self-compacting alkali-activated concrete, offering valuable guidance for optimizing activator dosage, water–binder ratio, and sustainability indices.

Although previous studies have examined alkali-activated systems incorporating FA, GGBFS, or RHA as individual precursors, a rigorous and unified investigation of Self-compacting geopolymer concrete (SCGC) formulated with a hybrid FA–GGBFS–RHA–micro-silica binder and activated using a neutral-grade, high-silica-modulus sodium silicate solution under ambient curing is missing from the existing literature. Furthermore, current studies do not provide a clear understanding of how the binder composition and solution-to-binder ratio interact to govern the rheology, strength development, and gel-phase evolution in such multi-component SCGC systems. This unexplored material–activator interaction constitutes the core research gap addressed in the present work.

Objectives of this study

- To optimize a multi-component hybrid binder system comprising FA, GGBFS, RHA, and micro-silica capable of achieving superior flow stability, cohesive self-compaction, and enhanced mechanical efficiency under practical ambient-curing conditions.
- Establishing a sustainability-centric mix-design framework that exploits the synergistic reactivity among supplementary cementitious materials, thereby reducing

the environmental footprint of concrete production while improving the structural integrity and long-term durability of SCGC.

- To benchmark the mechanical performance, including compressive, splitting tensile, and flexural strengths, of hybrid-binder SCGC activated with a neutral-grade, high-silica-modulus sodium silicate solution, and to quantify the resultant performance enhancements relative to conventional mixes.
- To elucidate the microstructural evolution of SCGC through a combined SEM, EDX, XRD, and FTIR characterization strategy, with the aim of identifying the gel morphology, phase assemblage, reaction products, and matrix densification mechanisms governing the strength development.
- To refine and validate a high-performance SCGC mix design by evaluating the influence of secondary parameters, such as the curing regime, aggregate characteristics, and functional additives, to formulate a robust, field-deployable optimization approach for ambient-cured geopolymer systems.

2 Materials

2.1 Materials used

The materials used in the production of (SCGC) are critical for defining their effectiveness and characteristics. The components used in the manufacturing of (SCGC) have a significant effect on the final quality. FA, GGBFS, RHA, and Micro Silica are used as binding materials in (SCGC), including natural river sand for fine aggregates, crushed granite for coarse aggregates, single activator solutions to replace water as an alkaline activator, and the addition of super plasticizers to improve the properties of the paste.

2.2 Binder materials

Fly ash (FA), ground granulated blast furnace slag (GGBFS), rice husk ash (RHA), and micro silica (MS) were used in this study. Class F fly ash was sourced from the Ramagundam Thermal Power Plant in India. Class F FA typically reduce the heat of hydration and water demand, and enhance resistance to sulfate attacks and chloride penetration. GGBFS, RHA, and MS were procured from Astrra Chemical Solutions (Chennai, India). GGBFS, obtained by rapidly cooling molten iron slag, contains silica, alumina, magnesia, and CaO. Incorporating GGBFS into geopolymer systems based on FA accelerated the setting times and increased the compressive strength under ambient curing. A higher GGBFS content further enhanced the strength properties at room temperature. RHA, with finer and spherical particles than ordinary cement, improves particle packing and concrete rheology. It acts as a partial replacement for alumino-silicate materials, reducing the dependency on natural resources. Owing to its high silica content, RHA enhances geopolymerization and improves the strength and durability of concrete. It also allows concrete to flow easily without vibrations, effectively filling complex shapes and densely reinforced areas. Micro silica (silica fume), a by-product of silicon alloy manufacturing, features extremely fine particles with a large surface area. These particles fill the concrete voids, increasing the density and improving the adhesion between the geopolymer binder and aggregates. This creates a robust interfacial transition zone, which significantly reduces potential deboning issues. The specific gravities for FA, GGBFS, RHA, and MS are 2.25, 2.83, 2.14, and 2.64, respectively [32]. The chemical compositions of the

Table 1 Fly ash, GGBFS, RHA and micro silica chemistry (as a mass)

Structures of chemicals	Fly-Ash (%)	GGBFS (%)	RHA. (%)	Micro silica
SiO ₂	58	34.07	92.30	99.920
Al ₂ O ₃	26.62	20	0.4	0.031
CaO	1.75	32.5	0.700	NILL
Fe ₂ O ₃	3.58	0.8	0.450	0.0012
SO ₃	1.8	0.9	0.045	0.2
Na ₂ O ₃	2	NIL	0.700	0.004
MgO	1.91	7.89	0.850	0.5
Loss on ignition	2	NIL	3.150	0.001

Table 2 Particle size and surface area of raw materials

Material	d ₅₀ (μm)	BET Surface Area (m ² /g)	Amorphous Silica (%)
GGBFS	12–14	0.45–0.65	28–38
Fly Ash (Class F)	18–22	0.30–0.40	45–60
Micro-silica	0.15–0.20	15–30	> 95
RHA	6–8	18–22	85–92

ground-granulated blast furnace slag, fly ash, rice husk ash, and micro-silica are listed in Table 1.

2.2.1 Particle size and surface area characterization

The particle size distributions of aluminosilicate and supplementary cementitious materials were determined using laser diffraction analysis, and the measured physical characteristics are summarized in Table 2. The median particle sizes (d_{50}) were recorded as 12–14 μm for GGBFS, 18–22 μm for fly ash, 0.15–0.20 μm for micro-silica, and 6–8 μm for rice husk ash (RHA). The corresponding BET-specific surface areas were 0.45–0.65 m²/g for GGBFS, 0.30–0.40 m²/g for fly ash, 15–30 m²/g for micro-silica, and 18–22 m²/g for RHA. The amorphous silica contents of the materials are 28–38% for GGBFS, 45–60% for fly ash, > 95% for micro-silica, and 85–92% for RHA. The combination of ultra-fine particle size, particularly in the case of micro-silica, together with a higher surface area and high amorphous silica proportion, enhances the dissolution of reactive species and promotes the rapid formation of aluminosilicate gel. These characteristics govern the nucleation behavior, reaction kinetics, and ultimately, the geopolymerization efficiency of the binder system.

2.3 Aggregates

Natural river sand, sieved through a 4.75 mm sieve, was used as a fine aggregate in SCGC, with a saturated surface dry specific gravity of 2.64, meeting Zone II criteria (IS-383, 2016). It enhances workability, flow-ability, and self-compaction, while its smooth, rounded particles improve cohesiveness and prevent segregation. River sand also strengthens SCGC, increasing the compressive strength, flexural strength, and crack resistance owing to strong particle interlocking. Crushed granite, with a maximum size of 20 mm and a specific gravity of 2.72, was used as the coarse aggregate. Its angular shape provides friction for effective stress transfer and, improves strength and stability. In addition, crushed granite is abrasion-resistant, durable, and has good drainage properties. The fineness modulus of the fine aggregate was 2.54, whereas that of the coarse aggregate was 6.95, which is consistent with the specifications of IS 383 (2016) [33].

Table 3 Details of mix proportions

Mix design	Binder (kg/m³)	AAS/BINDER	Binder in percentage				Fine aggregate (kg/m³)	Coarse aggregate (kg/m³)	AAS (kg/m³)	SP (% binder)	Extra water (% binder)
			GGBFS (%)	FA (%)	RHA (%)	MS (%)					
M1	500	0.5	60	20	15	5	814.7	679.1	300	4%	12%
M2	500	0.5	60	20	10	10	814.7	679.1	300	4%	12%
M3	500	0.5	60	20	5	15	814.7	679.1	300	4%	12%
M4	500	0.6	60	20	15	5	814.7	679.1	360	4%	12%
M5	500	0.6	60	20	10	10	814.7	679.1	360	4%	12%
M6	500	0.6	60	20	5	15	814.7	679.1	360	4%	12%
M7	600	0.5	60	20	15	5	756.7	630.8	300	4%	12%
M8	600	0.5	60	20	10	10	756.7	630.8	300	4%	12%
M9	600	0.5	60	20	5	15	756.7	630.8	300	4%	12%
M10	600	0.6	60	20	15	5	756.7	630.8	360	4%	12%
M11	600	0.6	60	20	10	10	756.7	630.8	360	4%	12%
M12	600	0.6	60	20	5	15	756.7	630.8	360	4%	12%
S0	500	0.5	0	100	0	0	814.7	679.1	300	4%	12%
S30	500	0.5	30	70	0	0	814.7	679.1	300	4%	12%
S60	500	0.5	60	40	0	0	814.7	679.1	300	4%	12%
S100	500	0.5	100	0	0	0	814.7	679.1	300	4%	12%

2.4 Alkaline solution

The sodium silicate (water glass) activator used in this study was rigorously characterized to ensure chemical consistency, transparency, and reproducibility. Commercial sodium silicate solutions supplied by Kiran Global Chem Ltd., Chennai, India, were examined using supplier batch certificates as part of the activator verification.

In this investigation, the silica modulus (M_s) is explicitly defined on a mass basis and is; expressed as: $M_s = (\text{mass \% of SiO}_2) / (\text{mass \% of Na}_2\text{O})$.

The neutral-grade sodium silicate solution (Product Code: KG-SS-N) exhibited a verified composition of 24.88 wt% SiO_2 and 12.46 wt% Na_2O , corresponding to a mass-based silica modulus (M_s) of 2.92. For completeness of activator characterization, an alkaline-grade sodium silicate solution (Product Code: KG-SS-A) was also examined, which contained 28.98 wt% SiO_2 and 9.92 wt% Na_2O , yielding a mass-based M_s of 1.99. All compositional values were confirmed using the supplier-provided batch certificates.

It is explicitly stated that only neutral-grade sodium silicate solution ($M_s = 2.92$) was used in all mix proportions, experimental programs, and analyses reported in this manuscript. The alkaline-grade sodium silicate solution was not used experimentally and was included solely for comparative reference and completeness of the activator documentation.

The activator was used exactly as supplied, without dilution, and no NaOH or secondary alkali was added at any stage. Prior to mixing, the neutral-grade sodium silicate solution exhibited a pH range of 11.9–12.7, density of 1.48–1.52 g/mL, and viscosity of 380–420 cP at 25 °C, ensuring consistent chemical activation and reproducible geopolymerization across all mixtures [34].

This explicit clarification of the activator chemistry, silica modulus definition, and activator selection removes ambiguity and provides a robust and unambiguous basis for interpreting the geopolymerization mechanisms and performance characteristics reported in this study.

2.5 Superplasticizer

A polycarboxylate-ether (PCE) based superplasticizer (Astrra Hyperplastic PCE-885, Astrra Chemicals, Chennai, India) was used to enhance the flow ability of (SCGC) [35]. The dosage of the superplasticizer was fixed at 4% by mass of the total binder content, which was within the typical range recommended for self-compacting geopolymer systems. The product had a solid content of $30 \pm 1\%$, density of 1.08–1.10 g/mL, and pH of 6.5–7.5. All the mixtures were prepared using the same Batch No (batch No. SP-885/2025-02) to ensure consistency. SP was added directly to the activator solution prior to mixing to ensure uniform dispersion and avoid agglomeration. No SP was added as a percentage of the liquid activator; the 4% dosage strictly corresponded to the binder mass basis.

3 Experimental program

The experimental program was designed to ensure a rigorous, reproducible, and internally consistent evaluation of the fresh, mechanical, and microstructural behaviors of Self-compacting geopolymer concrete (SCGC). The investigation comprised a clearly defined main ambient-cured experimental series and supplementary parametric studies,

which were explicitly distinguished to avoid ambiguity in mix design, water addition, and curing interpretation.

The primary experimental matrix consisted of SCGC mixtures produced using a hybrid binder system composed of 60% ground-granulated blast furnace slag (GGBFS), 20% fly ash (FA), and micro silica (MS) partially replaced with rice husk ash (RHA) at 5%, 10%, and 15% of the total binder mass. In addition, four reference mixes (S0, S30, S60, and S100) were prepared with GGBFS-to-FA ratios of 0/100, 30/70, 60/40, and 100/0, respectively. To systematically examine the influence of binder richness and liquid availability, the experimental program incorporated two binder contents (500 and 600 kg/m³) and two activator-to-binder (AAS/B ratios (0.50 and 0.60), which were consistent with the mix proportions summarized in Table 3. A neutral-grade sodium silicate solution ($M_s = 2.92$) was used as the sole alkaline activator in all the mixtures, and no NaOH or secondary alkali source was introduced at any stage. A polycarboxylate-ether superplasticizer was incorporated at a constant dosage of 4% by mass of the binder.

Fine and coarse aggregates were maintained under saturated surface-dry (SSD) conditions to ensure the precise control of the effective liquid-to-binder ratio. Aggregate contents were proportioned as 814.7/756.7 kg/m³ (fine) and 679.1/630.8 kg/m³ (coarse) for the 500 and 600 kg/m³ binder series, respectively, as detailed in Table 2. All materials were batched by mass, and mixing was carried out in a pan mixer using dry-mixing binders and aggregates for 2 min, followed by gradual incorporation of the activator–superplasticizer blend over 45–60 s. Wet mixing was continued for 3–4 min until a homogeneous and cohesive SCGC matrix was obtained. Specimens were cast without vibration, demolded after 24 ± 1 h, and cured strictly under ambient laboratory conditions (27 ± 2 °C and $70 \pm 5\%$ relative humidity) for 3, 7, and 28 days. Unless explicitly stated otherwise, all the results and discussions in this manuscript refer to this ambient-cured main experimental series.

In the main experimental series, a fixed extra-water content of 12% by binder mass was used uniformly for all mixes. The term *extra water* refers explicitly to additional free water independent of the sodium silicate activator (e.g., 60 kg/m³ for a 500 kg/m³ binder). To isolate the influence of excess free water, a separate extra-water parametric study was conducted using water contents of 10%, 12%, 15%, and 20%, while keeping all other mix parameters constant. These mixes are not part of the main experimental dataset and are discussed separately.

Table 3 Compressive strength development in 3, 7, and 28 days

Mix design	3-days	7-days	28-days
M1	29.87	43.775	51.5
M2	35.09	51.425	60.5
M3	35.554	52.105	61.3
M4	29.29	42.925	50.5
M5	35.67	52.275	61.5
M6	36.192	53.04	62.4
M7	30.45	44.625	52.5
M8	36.25	53.125	62.5
M9	37.41	54.825	64.5
M10	31.03	45.475	53.5
M11	35.67	52.275	61.5
M12	36.87	53.975	63.5

In addition, a limited heat-curing parametric investigation was performed to evaluate the influence of thermal activation on strength development. The selected mixes were oven-cured at 60, 70, 80, and 90 °C for durations of 21, 48, 72, and 96 h, followed by conditioning under ambient laboratory conditions for 24 h prior to testing. These heat-cured results were clearly distinguished from the ambient-cured dataset and were used solely to assess the curing sensitivity.

Fresh properties, including slump flow, T_{50} time, J-ring, V-funnel, and L-box, were evaluated in accordance with EFNARC (2005) guidelines.

All the mechanical properties were evaluated using three replicate specimens per mix and curing age ($n = 3$). The results are reported as mean values, and standard deviations (SD) are provided for representative control, intermediate, and optimized mixes to quantify experimental variability. Error bars corresponding to ± 1 SD are included in all the relevant graphical plots. The observed strength enhancements (20–30%) significantly exceeded the measured scatter, confirming that the reported improvements were statistically meaningful. The compressive strength was measured on 150 mm cubes following IS 516 (Part 5/Sect. 1):2018 at a loading rate of 0.20 MPa/s. The split tensile strength was determined for 150 × 300 mm cylinders in accordance with ASTM C496/C496M-17, and the flexural strength was evaluated for 100 × 100 × 500 mm prisms following ASTM C1609/C1609M-19. All testing equipment was factory calibrated before use. Microstructural characterization using SEM, EDX, XRD, and FTIR was performed on the 28-day ambient-cured specimens to elucidate the gel morphology, elemental distribution, phase assemblages, and densification mechanisms.

3.1 Fresh properties

Crushed coarse aggregate, fine sand (SSD), fly ash, GGBFS, RHA, and micro-silica were dry-mixed for 2.5 min in a 100-L mixer. The premixed alkaline solution, superplasticizer, and additional water were then added, followed by 3 min of wet mixing and brief manual mixing to ensure uniformity. Fresh SCC was tested for slump flow, T_{50} , J-ring, V-funnel, and V_5 -minute flow in accordance with EFNARC. Only the test procedures are presented here; all interpretations and discussions of the workability results are provided solely in the Results and Discussion section. All the fresh qualities meet the EFNARC criteria, although the rheological properties increase the micro silica content by 5% to 15% [36]. Figure 1. depicts the mixing of the (SCGC). This section presents only the test procedures and the experimental protocol. All fresh-property results and interpretations are discussed exclusively in the Results and Discussion sections.

3.2 Findings of the T_{50} slump flow test

The slump-flow diameter measurement is the main parameter for determining the behavior of concrete mixtures. It effectively measures the amount of time it takes for a concrete mixture to spread to a specified diameter that is 500 mm. This value is known as the T_{50} cm slumped flow time, which indicates the relative viscosity, as well as a relative estimate of the unrestricted SCC mixture flow rate. A shorter time suggests a better flow capacity. A cone-shaped mold was filled with the SCGC mixture during the T_{50} cm slump flow test. The cone was lifted once the concrete spread and flowed freely into the mold. The T_{50} cm number is the time required for concrete to attain a spread diameter of 50 cm. It may determine the binder content range that allows for a sufficient rate of



Fig. 1 Mixing of (SCGC)



Fig. 2 T_{50} slump flow test

concrete flow and spread, allowing for easy placement and form filling without excessive segregation or bleeding. According to the European Federation of National Associations for Concrete (EFNARC) procedure, a shorter T_{50} cm duration indicates a higher degree of flow ability in the concrete mixture. Figure 2. depicts the T_{50} cm slump flow test.

The T_{50} cm slump flow durations for these mixes ranged from 3 s to 4.2 s. Significantly, these numbers are within the permitted range outlined by the EFNARC recommendations, which state that the slump times for optimal performance should be between 2

and 5 s. Micro silica is well known for its small particle size and high reactivity with various components of concrete mixtures. The measured T_{50} slump-flow times ranged from 3.0 to 4.2 s, which lie within the EFNARC-recommended range of 2–5 s, confirming satisfactory self-compacting performance for all mixes. The combined effects of the solution-to-binder ratio, micro-silica dosage, and rice husk ash (RHA) content govern variations in T_{50} time. Micro-silica contributes to improved cohesiveness and flow stability only within the optimum dosage range (up to approximately 15%). In contrast, increasing RHA content increases T_{50} time due to its high specific surface area and porous morphology, which increases water demand and mixture viscosity. The test findings reveal the rate at which the concrete flows and spreads, showing the ease of placement and the capacity of the concrete to fill the formwork without additional compaction. The T_{50} cm slump flow test can also be used to evaluate the SCGC flow ability and rejection of segregation. Higher T_{50} cm values may indicate improved flow ability and lower segregation risk, indicating that the concrete mixture can maintain homogeneity and uniformity during placement [37].

3.3 Slump flow results

Slump flow tests are the most basic and popular methods for determining the flow ability of SCC. This test provides a precise evaluation of the filling capacity and establishes the SCC flow rate in the absence of obstructions. The basic tools used in this test were the same as those used in the traditional slump test. The only difference was that rodded concrete was used instead of filled molds. The slump cone finished testing when it was set up on a hard, non-absorbent leveling plate and filled with concrete without being compacted. Following filling, the slump cone was raised vertically to facilitate the unrestricted flow of the concrete. Two perpendicular measurements of the concrete diameter were performed, and the average of the two diameters was recorded.

The test results for the slump flow of several SCGC mixes are shown in Fig. 3. The experimental results clearly indicate that increasing rice husk ash (RHA) content reduces slump flow. This reduction is attributed to the high specific surface area and internal porosity of RHA, which increases the water demand and limits the availability of free activator solution for flow. Accordingly, RHA did not enhance workability, and this trend has been consistently reported throughout the manuscript. Concrete is deemed to have sufficient filling capability when its slump flow values fall between 600 and 800 mm, according to the EFNARC standards. M_1 – M_{12} mixes were prepared with a stable binder content of 500 kg/m³ or 600 kg/m³, various proportions of solutions/binders (0.5, 0.6), and a consistent activation solution of Ms 2.92. M_1 ($G_{60}F_{20}R_{15}M_5$), which is the reference mix, had a slump flow of 618 mm. Mix M_{12} ($G_{60}F_{20}R_5M_{15}$) had a significantly higher slump flow of 717 mm, and the slump flow values increased continuously as the solution-to-binder ratio increased, as did the incorporation of various amounts of micro-silica (from 5% to 15%). Furthermore, all mixes that included RHA instead of micro silica had higher slump flow values. This aids in the optimization of combination proportions for self-compaction and desired fresh qualities [38]–[39].

3.4 Results of the J-ring test

The slump cone was positioned in the middle of the J-ring and filled with concrete to conduct this test in tandem with the slump flow test. Concrete can spread horizontally



Fig. 3 Slump flow

through the spaces between the bars when the slump cone is lifted. Similar to the slump flow test, the spread diameter and T_{50} time were also noted. The horizontal spread was then compared with and without the J-Ring. Furthermore, four measurements were made of the height differential between the concrete exterior and interior of the J-Ring. The variation between J-ring and slump flows provides a blocking evaluation and indicates the extent to which a certain combination may pass. Good passing ability is indicated by a difference of less than 25 mm, and poor passing ability is indicated by a variance of more than 50 mm. To assess the flow, we used the J-ring test and the possible properties of the concrete mixture; the results are shown in Fig. 4. According to the EFNARC recommendations from 2002, a good range for the off-blocking step is specified as 0–10 mm. Visual evaluations and documented data suggested that all 12 mixes demonstrated commendable passing ability.

Furthermore, the J-ring measurements for several SCGC blends were within the permitted range of 0–10 mm, according to the EFNARC recommendations. In view of these findings, it is worth noting that numerous (SCGC) mixtures were J-ring tested using a single activator solution containing 2.92 Micro Silica (MS). These tests produced values ranging from 8 mm to 9.4 mm, with a minimum J-ring value of 8 mm obtained for mix M_{12} using a single activator solution with an MS concentration of 2.92. Under identical conditions, the maximum J-ring value for M_1 was 9.4 mm. Within the investigated range, the optimized micro-silica dosage contributed to improved cohesiveness and passing ability; however, increasing the RHA content resulted in slightly higher J-ring values owing to increased mixture viscosity, while remaining within the EFNARC limits.



Fig. 4 J-ring test

The J-ring test identifies the ideal binder content for obtaining good passing ability and resistance to segregation in SCGC and provides insights into the ability of concrete to maintain its homogeneity and consistency throughout placement [40].

3.5 V-funnel test results

This test is generally used to assess the SCC filling ability, but it can also be used to assess the segregation resistance. The V-funnel diameters varied; however, the bottom hole was often 65 mm x 75 mm or 75 mm x 75 mm. Approximately 12 L (0.4 ft³) of concrete was required for this test, and the concrete had to fill the funnel completely without being tapped or compacted. The V-funnel flow time is the amount of time it takes for the concrete to completely pass through the orifice after the funnel has been filled with concrete, and the bottom trap door is opened to allow the concrete to flow out naturally. This test was designed to assess the capacity of the SCC to flow through a continuously diminishing portion without segregation or blockage. The SCC normally prefers a funnel flow time of 6–12 s. The V-funnel test was used to assess the stability and flow properties of the SCGC mixes activated with single activator solutions with a molecular weight of 2.92. Figures 5 and 6 shows the results for both V-funnel flow time and V-funnel T_5 (flow time at 5 min), in accordance with the 2002 EFNARC criteria.

According to the findings, when a single activator solution of Ms 2.92 was used, the V-Funnel values varied between the three (SCGC) mixtures, ranging from 7.1 s to 9.2 s. These values were within the EFNARC-recommended allowed range of 6–12 s. Mix



Fig. 5 V-funnel test

M₁₂ had the shortest recorded V-funnel flow time of 7.1 s, whereas Mix M₁ had the longest recorded V-funnel flow time of 9.2 s. The observed V-funnel flow times reflect the combined influence of the activator content, micro-silica dosage, and RHA replacement. While the optimized micro-silica dosage supported flow stability, increasing the RHA content increases the flow time owing to the higher water demand. The flow time indicates the fluidity and passage capability of a concrete mixture. A shorter flow time suggests an improved flow ability and pass ability. The same pattern of behavior was observed with the V-funnel T₅ (flow time of 5 min) [41].

3.6 Hardened properties

The compressive strength development of SCGC is governed by the interplay between the curing temperature, calcium availability, and dissolution rate of aluminosilicate precursors. Increasing the curing temperature from 60 °C to 70 °C enhanced the dissolution of Ca-, Si-, and Al-bearing species from GGBFS and FA, accelerating condensation reactions and promoting the early formation of hybrid C–A–S–H and N–A–S–H gels. This results in a higher early age strength. However, curing above ~70 °C offers diminishing gains, as most polymerization has already occurred, and excessive thermal exposure can induce coarser pore networks, leading to minor reductions in strength.

Because of the slower dissolution rate of the aluminum and silicon monomers, the mixes had inferior toughness qualities at ambient temperature when compared to specimens cured at 70 °C and 60 °C. The presence of hydroxide ions increased the reactivity of



Fig. 6 V 5 minutes funnel test

GGBFS. This mechanism successfully aids in the breakdown of chemical bonds present in the Si-O-, Ca-O, and Al-O structures, leading to the creation of dissolved species such as $[\text{H}_3\text{SiO}_4]$, Ca_2 , $[\text{H}_2\text{SiO}_4]_2$, and $[\text{Al}(\text{OH})_4]$. It is important to clarify that the ionic species such as $[\text{H}_3\text{SiO}_4]^-$, Ca^{2+} , $[\text{H}_2\text{SiO}_4]^{2-}$, and $[\text{Al}(\text{OH})_4]^-$ mentioned in this section were not directly measured or quantified in the present work. Their inclusion is based on well-established dissolution–polymerization mechanisms reported in the geopolymer chemistry literature, wherein hydroxide ions initiate the depolymerization of aluminosilicate networks and release monomeric silicate and aluminate species into the pore solution. These theoretical species constitute the commonly accepted reaction pathway for alkali-activated systems, but their presence in this study was inferred rather than experimentally verified. Accordingly, the manuscript has been revised to explicitly state that these ions represent literature-supported mechanistic descriptions, rather than the outcomes of direct chemical analysis. This chemical reaction produces C-S-H, sodium alumina silicate hydrate (N-A-S-H) gel, calcium aluminum silica hydrate (C-A-S-H) gel, and a, resulting in a denser microstructure. It is important to clarify that the identification of C-S-H, N-A-S-H, and C-A-S-H gels in this study was based on indirect microstructural indicators rather than direct chemical fingerprinting. The coexistence of these gels can be inferred from the following : (i) SEM observations showing continuous amorphous gel matrices and reduced void connectivity; (ii) EDX elemental ratios indicating

Ca-rich and Na-balanced aluminosilicate domains; (iii) broadened amorphous halo in XRD, characteristic of disordered aluminosilicate and Ca-modified phases; and (iv) red-shift of the Si–O–T stretching band in FTIR, reflecting increased polymerization and Ca incorporation. These four lines of evidence together provide strong but indirect confirmation of a hybrid C–A–S–H/N–A–S–H gel system, consistent with the widely accepted reaction mechanisms in alkali-activated materials. Compared with Fly Ash (FA), GGBFS contains more calcium dioxide (CaO) and higher quantities of silica and alumina. The major constituents of FA are crystalline alumina and silica. Simultaneously, GGBFS exhibited a higher degree of amorphousness, which increased its reactivity. The addition of calcium oxide (CaO) to GGBFS accelerates the strength development process of the GP paste, thereby improving the early stage strength. The rapid reaction involving the increased calcium content in GGBFS can be attributed to the accelerated attainment of a high starting strength in mixtures containing GGBFS. When GGBFS is mixed with water, an exothermic reaction involving calcium oxide (CaO) occurs, resulting in the generation of heat. Even at room temperature, this heat stimulated the initiation of geopolymerization in the combination. The geopolymer paste containing 50% GGBFS exhibited a higher density and lower porosity than the paste containing 100% GGBFS. This property significantly contributes to the strength of the geopolymer compositions. The addition of GGBFS to the mixture increased the quantity of the calcium compounds. This promoted the production of a larger C–S–H gel. The use of GGBFS in the geopolymerization process resulted in the formation of a composite material composed of C–S–H gel. This gel is formed by a chemical reaction between CaO and aluminosilicate gel, which is produced by SiO₂ dissolution, resulting in the successful production of a geopolymer, specifically C–A–S–H hydrate. The presence of C–A–S–H in the gel causes the solidification and early strengthening of the SCGC material, which is predominantly composed of GGBFS. In the present study, RHA influenced the mechanical response primarily by modifying the gel chemistry and pore structure, as evidenced by the strength development and microstructural observations. The incorporation of RHA into geopolymer concrete changes its physical properties and reaction processes, resulting in major changes in the geopolymer matrix.

The SiO₂/Al₂O₃ ratio of the source material can be efficiently adjusted and maintained using reactive hydrous alumina with a high SiO₂ concentration. Geopolymers with higher silica alumina ratios were more flexible. The presence of excess H₂O in the geopolymer matrix resulted in void formation. The porous structure of the RHA particles improved pore dispersion and increased the weight of the geopolymer matrices, allowing fine particles to effectively fill these spaces. Consequently, the mechanical properties of SCGC are enhanced when recycled high alumina is used in geopolymer composites, which enhances the compressive strength and reduces the pore width by up to 55%. After ensuring the essential workability requirements as directed by EFNARC, the fresh concrete mixtures were stored in an oven at a set temperature for a specified duration based on the chosen test variables. The test specimens were detached from the oven at the end of the curing period and kept undisturbed for approximately 15 min to avoid a sudden shift in the ambient conditions. After demolding the test specimens and allowing them to air-dry at room temperature for 24 h, they were evaluated for CS [42–44].

3.6.1 Compressive strength

One of the most notable features of hardened concrete is its compressive strength, which is used as a characteristic material value in concrete classification. Table 3 shows the compressive strength results for the self-compacted geopolymer concrete with various mix proportions after three, seven, and 28 days of testing. All the combinations had the same composition of GGBS (60%) and fly ash (20%). However, the percentage of micro silica (MS) substituted with RHA varies among mix designs, ranging from 5 to 15%. A single activator solution was used to activate these combinations. The strength of compression of self-compacted GPC mixes activated using a single activator solution with Ms 2.92 varies according to the mix composition. The compressive strength after three days of curing ranges from 37.41 to 29.29 MPa for the M₁ through M₁₂ mix compositions. Similarly, the compression strength varied from 54.825 to 42.925 MPa after seven days of curing. Finally, the strength of compression ranged from 64.5 to 50.5 MPa following a 28-day curing period. These data demonstrate the major impact of the micro silica replacement percentage with rice husk ash on the strength of the compression development of self-compacted GPC. The changes discovered at various curing times provided useful insights into the performance of these concrete mixtures and their possible use in diverse construction projects. The findings imply that by carefully selecting the mix amounts and curing durations, concrete with tailored compressive strength characteristics suitable for specific engineering needs can be produced. Further research and testing may yield new insights into refining these mixed designs for practical construction applications [45].

All the mechanical properties reported in this study represent the mean of $n = 3$ replicate specimens for each mix and curing age. Standard deviations (SD) are provided in all tables and error bars (± 1 SD) are included in all graphical representations. The observed improvements, such as the 26% compressive strength gain between the optimized and control mixes, exceeded the measured experimental scatter and therefore represented statistically meaningful differences. This statistical framework ensures that the mechanical trends reported herein are reproducible, robust, and not artifacts of specimen-level variability. Table 4 shows compressive strength of SCGC is affected by the water addition, curing time, and curing temperature. Although extra water improved workability, excessive water caused segregation and bleeding, leading to a clear reduction in strength, which is consistent with the known effect of higher water-to-binder ratios [45]. The strength increased with longer curing times, with the maximum value observed at 96 h, whereas only minor improvement occurred after 48 h [46]. Higher curing temperatures also enhanced geopolymerization, and compressive strength increased as the curing temperature increased from 60 to 90 °C, confirming the importance of thermal activation in SCGC systems [47].

3.6.2 Split tensile strength

The split tensile strength of Self-compacting geopolymer concrete (SCGC) was evaluated at 28 days using cylindrical specimens of 150 mm diameter and 300 mm height, in accordance with ASTM C496/C496M-17. All results reported in this subsection correspond exclusively to the experimental program of the present study.

The 28-day split tensile strength values ranged from 2.30 to 3.46 MPa, depending on the binder content and mix composition. Mixes with a higher binder content of 600 kg/

Table 4 SCGC compressive strength with extra water, curing, and temperature

Mix conditions	Binder (kg/m ³)	GGBFS (%)	FA (%)	AAS/B (%)	Extra water (%)	Curing temp (°C)	Curing duration	Curing regime	28-CS(MPa)
Extra Water 10%	500	60	40	0.5	10	70	48 h + ambient	Heat-cured	53
Extra Water 12%	500	60	40	0.5	12	70	48 h + ambient	Heat-cured	46
Extra Water 15%	500	60	40	0.5	15	70	48 h + ambient	Heat-cured	36
Extra Water 20%	500	60	40	0.5	20	70	48 h + ambient	Heat-cured	23
Curing Time 21 h	500	60	40	0.5	12	70	21 h + ambient	Heat-cured	45
Curing Time 48 h	500	60	40	0.5	12	70	48 h + ambient	Heat-cured	51
Curing Time 72 h	500	60	40	0.5	12	70	72 h + ambient	Heat-cured	50
Curing Time 96 h	500	60	40	0.5	12	70	96 h + ambient	Heat-cured	51
Temperature 60 °C	500	60	40	0.5	12	60	48 h + ambient	Heat-cured	46
Temperature 70 °C	500	60	40	0.5	12	70	48 h + ambient	Heat-cured	52
Temperature 80 °C	500	60	40	0.5	12	80	48 h + ambient	Heat-cured	49

m³ consistently exhibited a higher split tensile strength than those with 500 kg/m³, indicating improved tensile resistance. This enhancement was attributed to better matrix densification and stronger bonding between the geopolymer binder and aggregates under ambient curing conditions.

It is explicitly stated that the present experimental program investigated only solution-to-binder ratios of 0.50 and 0.60, as defined in the mix design matrix. Accordingly, all the discussions and interpretations presented in this subsection are strictly limited to these experimentally investigated ratios.

3.6.3 Flexural strength

The flexural strength of SCGC was experimentally determined at 28 d using prism specimens of 100 × 100 × 500 mm, tested under third-point loading in accordance with ASTM C1609/C1609M-19. All the flexural strength results reported herein were derived solely from the present experimental investigation.

The results indicate that the flexural strength increases with higher binder content and optimized RHA–micro-silica replacement levels. Mixes containing 600 kg/m³ binder exhibited superior flexural performance compared to the corresponding mixes with 500 kg/m³ binder, reflecting improved crack resistance and stress transfer capacity. This behavior is attributed to the formation of a denser and more homogeneous geopolymer matrix, which is consistent with the observed microstructural characteristics.

The flexural strength results reported in this study represent the material-level performance obtained from laboratory-scale prism specimens and should not be directly extrapolated to large-scale structural behavior.

3.6.4 Comparison with previous studies

To contextualize the experimental findings of the present study, we selected insights from prior research on self-compacting geopolymer concrete are briefly highlighted. Previous studies [48, 49] have reported that lower solution-to-binder ratios (e.g., 0.35–0.45) enhance the mechanical performance by increasing the paste density and reducing the free water content. Further investigations [50, 51] demonstrated that wider variations in slag and fly ash contents significantly influence both the fresh and hardened properties of geopolymer systems, and that large-scale self-compacting geopolymer concrete beams exhibit flexural behavior and failure modes comparable to those of conventional reinforced concrete beams, particularly at higher binder contents and lower solution-to-binder ratios (0.70).

All observations discussed in this section are entirely literature-based and are included solely for comparison and contextual understanding. The present experimental program did not investigate solution-to-binder ratios of 0.35, 0.45, or 0.70, slag or fly ash contents beyond those defined in the mix design matrix, or large-scale flexural beam behavior. Accordingly, these findings should not be interpreted as outcomes of the present study.

3.6.5 Numerical consistency statement

All the numerical values reported in this manuscript have been cross-verified with the corresponding tables and Figures. The percentage improvements were calculated using the following expression:

$$\% \text{ Gain} = ((f_c(\text{mix}) - f_c(\text{control})) / f_c(\text{control})) \times 100.$$

For illustration, the 28-day compressive strengths of the control mix (M1:-51.5 MPa) and optimized mix (M9:-64.5 MPa) corresponded to a verified strength enhancement of 25.2%. All early age strength ratios, maximum strength values, and percentage gains strictly matched the data presented in the corresponding tables and Figures. The absolute strength values are reported exclusively in MPa, whereas the percentage values are dimensionless.

4 Result with discussion

Fresh property tests, including slump-flow, T_{50} time, V-funnel, J-ring, and L-box, were conducted in accordance with EFNARC (2005) to evaluate the flow ability and passing ability of SCGC mixtures. This study investigated the effects of binder content and solution-to-binder ratio on various self-compacted geopolymer concrete compositions. Furthermore, this study examines how different binder formulations with single activator solutions affect the fresh and hardened characteristics, as well as the microstructural qualities, of self-compacting GPC. The study began by emphasizing the significance of optimizing the mix proportions in geopolymer concrete, which is known for its environmentally friendly and sustainable features. Because of its capacity to flow and fill molds without vibration or external compaction, self-compacting GPC is a particularly interesting version that can greatly increase construction efficiency. A variety of binder contents and solution-to-binder ratios were carefully evaluated throughout the investigation to determine their impact on the characteristics of the concrete and an examination of fresh concrete properties such as workability and flow-ability, which are crucial for guaranteeing accurate placement throughout construction. A microstructural analysis was also included in this study to gain insights into the internal structure of concrete,

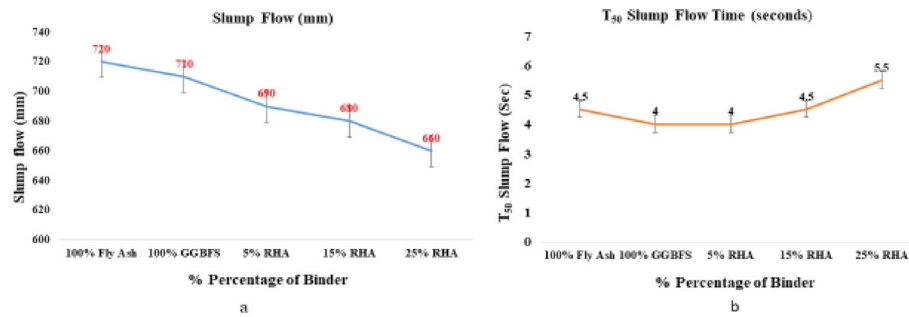


Fig. 7 Analysis of effect of RHA on (a) slump flow (b) T_{50} cm slump flow

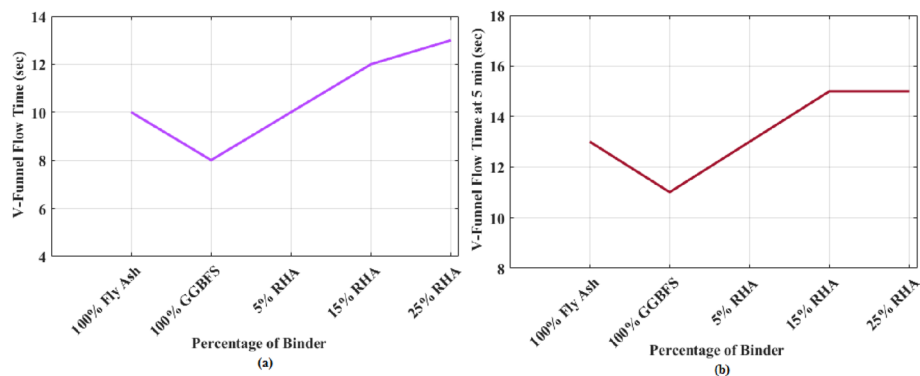


Fig. 8 Analysis of effect of RHA on (a) V-funnel flow (b) V_{5min} flow

which can provide useful information about its composition and potential areas for development. Finally, this work adds to the expanding body of information on self-compacted geopolymer concrete by thoroughly investigating the mix proportions and their impact on various properties. These findings are important not only for scholars, but also for engineers and practitioners who want to use geopolymer concrete in sustainable construction.

Figure 7 depicts the analyses of the effect of RHA on (a) slump flow and (b) T_{50} cm slump flow. Here, 7(a) shows the analysis of the effect of RHA on slump flow, in which, for 100% FA, the slump flow attains 720 mm for 100% GGBFS, the slump flow attains 710 mm, for 5% RHA, the slump flow attains 690 mm, for 15% RHA, the slump flow attains 680 mm, for 25% RHA, and 655, mm. Further 7(b) shows the analysis of effect of RHA on T_{50} cm slump flow in which, for 100% of FA the T_{50} cm slump flow attains 4.5 mm, for 100% of GGBFS the slump flow attains 4 mm, for 5% of RHA the slump flow attains 4 mm, for 15% of RHA the slump flow attains 4.5 mm and for 25% of RHA the slump flow attains 5.5 mm respectively.

Figure 8 depicts the analysis of the effect of RHA on the (a) V-Funnel Flow and (b) V_{5min} Flow. Here, 8(a) shows the analysis of the effect of RHA on the V-funnel flow in which, for 100% FA, the flow of a V-shaped funnel attains a time of 10s, for 100% GGBFS, the flow of a V- shaped funnel attains a time of 8s, for 5% RHA, the V-funnel flow attains a time of 10s, for 15% RHA, the flow of a V-shaped funnel attains a time of 12s and for 25% RHA, the V-funnel flow attains a time of 13s respectively. Figure 8(b) shows the analysis of the effect of RHA on V_{5min} flow in which, for 100% FA, V_{5min} attains a time of 13s, for 100% GGBFS, V_5 min attains a time of 11s, for 5% RHA, V_{5min} attains a time of

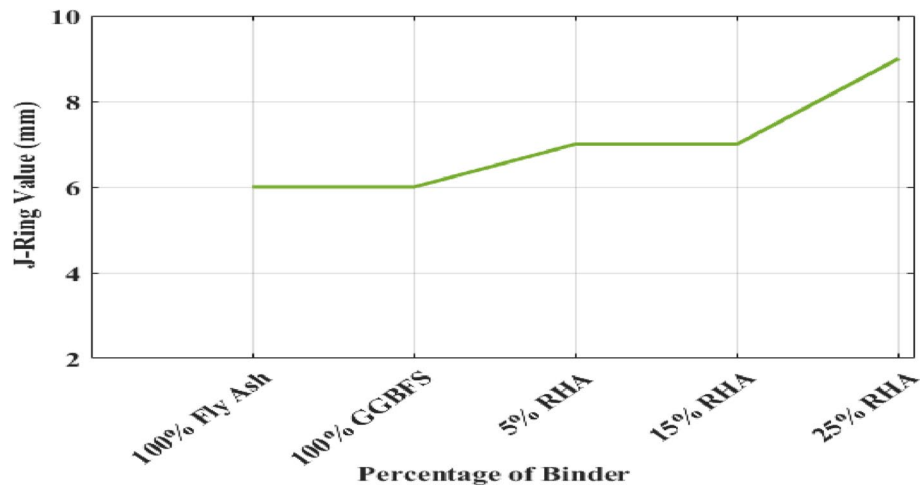


Fig. 9 Analysis of effect of RHA on J-ring value

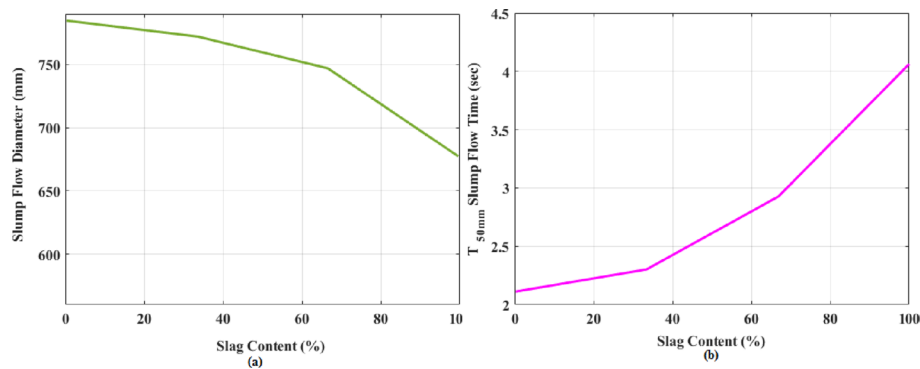


Fig. 10 Analysis of influence of slag content on (a) slump flow (b) $T_{50\text{mm}}$ flow time

13s, for 15% RHA, $V_{5\text{min}}$ attains a time of 15s and for 25% RHA, $V_{5\text{min}}$ attains a time of 15s respectively.

Figure 9 depicts the analysis of the effect of RHA on the J-ring value. Here, for 100% FA, the J-ring value attains 6 mm; for 100% GGBFS, the J-ring value attains 6 mm; for 5% RHA, the J-ring value attains 7 mm; for 15% RHA, the J-ring value attains 7 mm; and for 25% RHA, the J-ring value is 9 mm.

Figure 10 depicts the analysis of the influence of slag content on (a) Slump Flow and (b) $T_{50\text{mm}}$ Flow Time. Here, Fig.10(a) shows the analysis of the influence of slag content on the slump flow, where the slump flow diameter attains a lower of 790, 780, 770, 750, 720, and 680 mm at slag contents of 0%, 20%, 40%, 60%, 80%, and 100%, respectively. Further 10(b) shows the analysis of influence of slag content on the $T_{50\text{mm}}$ flow time where the $T_{50\text{mm}}$ slump flow attains a higher time of 2.1s, 2.2s, 2.4s, 2.8s, 3.4s and 4.1s at the slag content of 0%, 20%, 40%, 60%, 80% and 100% respectively.

Figure 11 depicts the analysis of the influence of slag content on (a) J-Ring Flow and (b) J-Ring Passing Ability. Figure 11(a) shows the analysis of the influence of slag content on the J-ring flow when the slump flow diameter attains lower values of 790, 780, 770, 750, 720, and 680 mm at slag contents of 0%, 20%, 40%, 60%, 80%, and 100%, respectively. The 11(b) shows the analysis of influence of slag content on the J-ring passing ability where

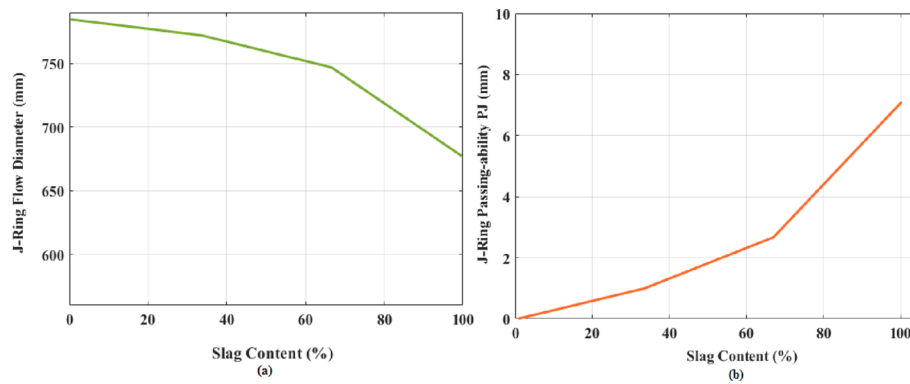


Fig. 11 Analysis of influence of slag content on (a) J-ring flow (b) J-ring passing ability

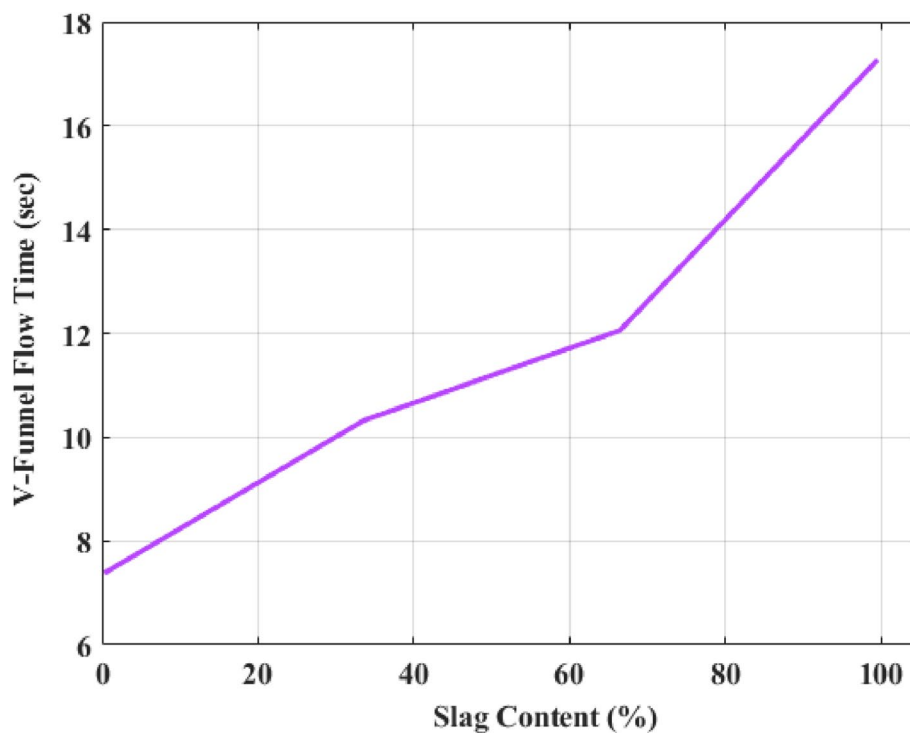


Fig. 12 Analysis of influence of slag content on the V-funnel flow time

the J-ring passing ability attains a higher of 0 mm, 0.6 mm, 1.2 mm, 2.4 mm, 4.4 mm and 7 mm at the slag content of 0%, 20%, 40%, 60%, 80% and 100% respectively.

Figure 12 depicts the analysis of the influence of slag content on the V-funnel flow time. Here, when the flow of a V-shaped funnel attains a higher time of 7.6s, 9s, 10.8s, 11.8s, 14s and 17.2s at the slag content of 0%, 20%, 40%, 60%, 80% and 100% respectively.

The reduction in the slump flow and the corresponding increase in T_{50} slump-flow time with increasing RHA content can be fully justified based on the physico-chemical characteristics of RHA. RHA has a very high specific surface area and a highly porous microstructure, which results in greater absorption of mixing water and a significant decrease in the free water available for flow. Therefore, mixes containing higher RHA inherently exhibit a lower slump and higher T_{50} times. This behavior is consistent with existing studies on RHA-blended SCC and geopolymer systems. The reduction in slump

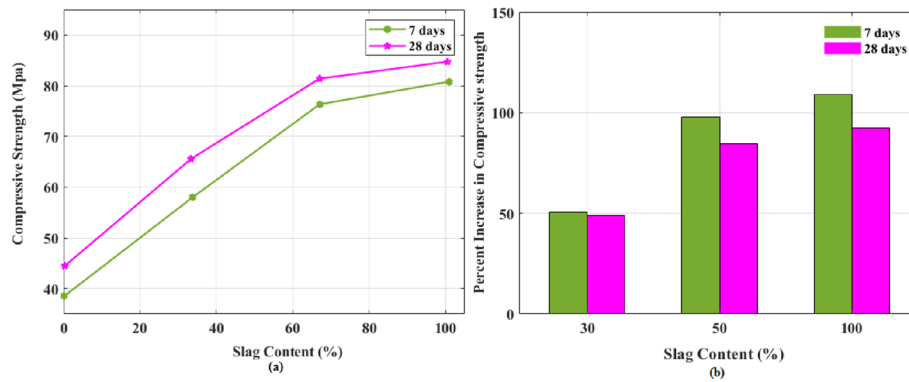


Fig. 13 Analysis of compressive strength at 7 and 28 days (a) influence of slag content (b) percent increase

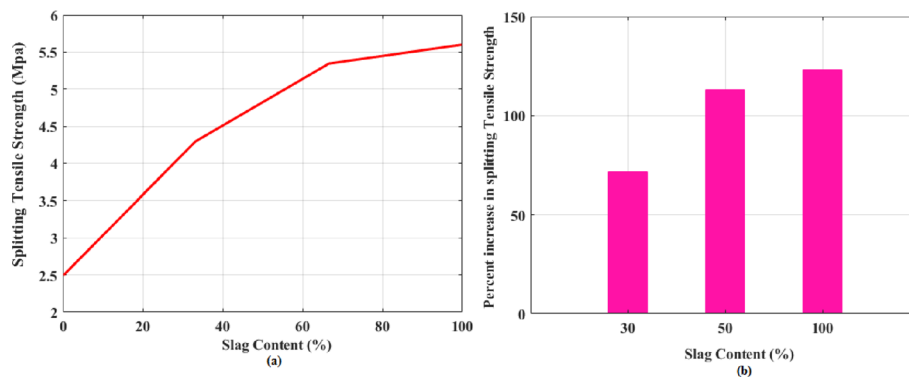


Fig. 14 Analysis of split tensile strength (a) influence of slag content (b) percent increase

flow and the corresponding increase in T_{50} time observed with higher RHA contents are fully consistent with trends widely reported in the literature, where the elevated water demand of RHA governs the fresh-state behavior [52–54].

The absolute compressive and splitting tensile strengths reported in this study were expressed in MPa. The quantities shown in Fig. 13(b) and Fig. 14(b) are percentage increases and are therefore dimensionless (expressed as %). Any previous text that inadvertently combined percent increase with MPa has been corrected: percentage improvements are unitless and reported as percentages, while MPa is used exclusively for absolute strength values.

Figure 13 depicts the analysis of the compressive strength at 7 and 28 d. (a) Influence of Slag Content (b) Percentage increase. Figure 13(a) shows the analysis of the influence of slag content on the CS at 7 and 28 days, in which 7 days, the compressive strength attains higher of 38 MPa, 58 MPa, 76 MPa and 80 MPa at slag contents of 0%, 36%, 68%, and 100%, respectively. In addition, after 28 d, the compressive strengths were higher at 45 MPa, 65 MPa, 82 MPa and 85 MPa at slag contents of 0%, 36%, 68%, and 100%, respectively. Moreover, Fig. 13(b) shows the analysis of the percent increase in compressive strength at 7 and 28 days, in which 7 days, the percent increase in compressive strength attains a higher value of 50%, 99%, and 110% at slag contents of 30%, 50%, and 100%, respectively. For 28 d, the percentage increase in compressive strength attained higher values of 49%, 80%, and 90% at slag contents of 30%, 50%, and 100%, respectively [55].

Figure 14 depicts the analysis of the split tensile strength (a) Influence of Slag Content (b) percentage increase. Here, 14(a) shows the analysis of influence of slag content on the split tensile strength at 28 days when the splitting tensile strength attains a higher of 2.5 MPa, 3.5 MPa, 4.5 MPa, 5.2 MPa, 5.5 MPa and 5.6 MPa at the slag content of 0%, 20%, 40%, 60%, 80% and 100% respectively. Moreover, Fig. 14 (b) shows the analysis of the percent increase in split tensile strength, where the percent increase in split tensile strength attains a higher values of 70%, 110%, and 130% at slag contents of 30%, 50%, and 100%, respectively [56].

Figure 15 depicts the analysis of the effect on the compressive strength, (a) extra water, (b) curing time, and (c) curing temperature. Here, 15(a) shows the analysis of the effect of extra water on CS, where 10% of extra water added to SCGC it attains 54 MPa compressive strength of, also when 12% of extra water is added to SCGC it attains 46 MPa compressive strength, then 15% of extra water is added to SCGC it attains 38 MPa compressive strength, and when 20% of extra water is added to SCGC it attains 23 MPa compressive strength. Furthermore, Fig. 15(b) shows the analysis of the effect of extra water on the compressive strength, in which for 21 h of curing, the compressive strength of 45 MPa, for 48 h of curing, the SCGC attains a compressive strength of 52 MPa, for 72 h of curing, the SCGC attains a compressive strength of 51 MPa and for 96 h of curing, the SCGC attains a compressive strength of 52 MPa. Moreover, Fig. 15(c) shows the analysis of the effect of curing temperature on CS, in which for 60 °C of curing temperature the SCGC attains a compressive strength of 45 MPa, for 70 °C of curing temperature the SCGC attains a CS of 52 MPa, for 80 °C of curing temperature, the SCGC attains a CS of 48 MPa and for 90 °C of curing temperature the SCGC attains a CS of 47 MPa.

Figure 16 depicts the analysis of the correlation between (a) CS and TS for SCGCs and (b) CS and FS for SCGCs. Here, 16(a) shows the analysis of the correlation between CS and TS for SCGC, in which the compressive strength attains higher of 47 MPa, 50 MPa, 65 MPa, 68 MPa and 70 MPa at splitting tensile strength of 4.8 MPa, 5 MPa, 5.5 MPa, 6 MPa and 6.4 MPa respectively. Further 16(b) shows the analysis of the correlation between CS and flexural strength for SCGC, where the compressive strength attains

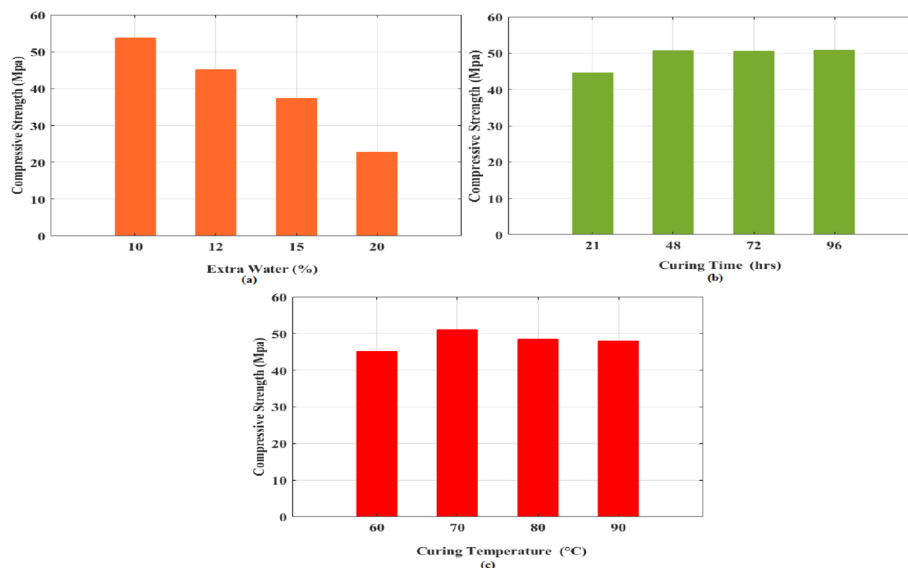


Fig. 15 Analysis of effect on compressive strength (a) extra water (b) curing time (c) curing temperature

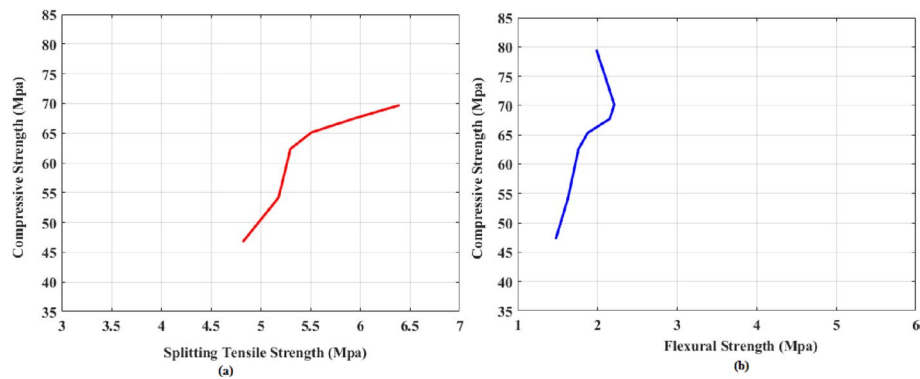


Fig. 16 Analysis of correlation between (a) CS and TS for SCGCs (b) CS and FS for SCGCs

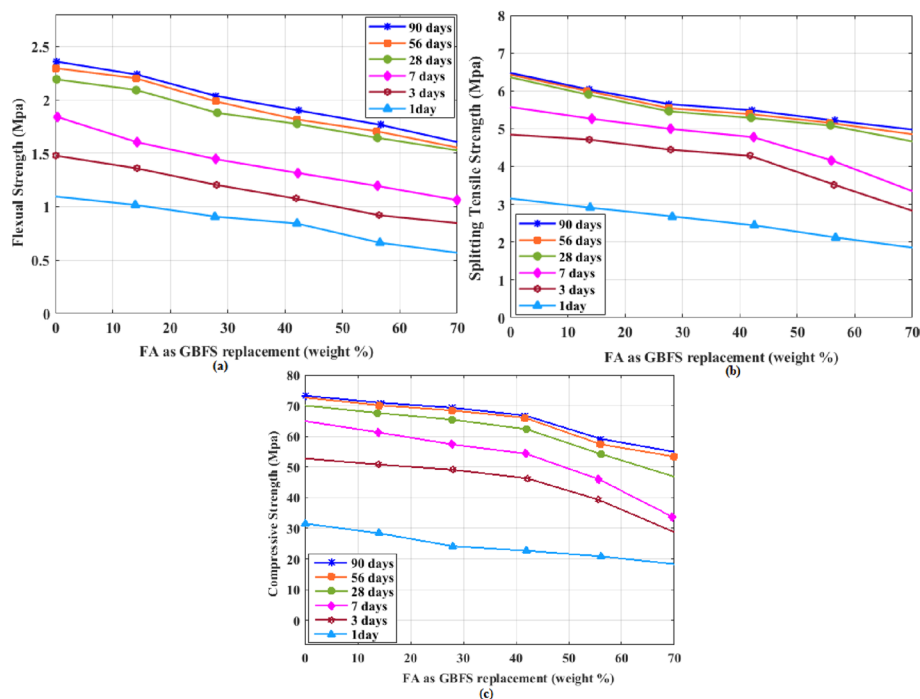


Fig. 17 Analysis of effect of FA content on (a) flexural strength development (b) split tensile strength development (c) compressive strength development

higher of 47 MPa, 55 MPa, 65 MPa, 70 MPa and 80 MPa at splitting tensile strength of 1.5 MPa, 1.7 MPa, 1.9 MPa, 2.2 MPa and 2 MPa respectively [57].

Figure 17 depicts the analysis of the FA content on (a) flexural strength development, (b) split tensile strength development, and (c) compressive strength development. Here, 17(a) shows the analysis of the effect of FA content on the flexural strength development of SCGC in which for 1 day the flexural strength of SCGC attains 1 MPa, 0.9, 0.8 MPa and 0.6 MPa when FA is replaced with GBFS by weights of 13%, 28%, 42%, and 56%, respectively. Then after 3 days, the flexural strength of SCGC attains 1.5 MPa, 1.4 MPa, 1.3 MPa, 1.2 MPa and 0.9 MPa when FA is replaced with GBFS by the weight of 0%, 13%, 28%, 42% and 56% respectively. Then after 7 days the flexural strength of SCGC attains 1.9 MPa, 1.6 MPa, 1.4 MPa, 1.3 MPa, 1.2 MPa and 1.1 MPa when FA is replaced with GBFS by the weight of 0%, 13%, 28%, 42%, 56% and 70% respectively. Also, after 28 days

the flexural strength of SCGC attains 2.2 MPa, 2.1 MPa, 1.9 MPa, 1.8 MPa and 1.7 MPa when FA is replaced with GBFS by the weight of 0%, 13%, 28%, 42% and 56% respectively. Further after 56 days the flexural strength of SCGC attains 2.3 MPa, 2.2 MPa, 2 MPa, 1.8 MPa and 1.7 MPa when fly ash (FA) is replaced with GBFS by the weight of 0%, 13%, 28%, 42% and 56% respectively. Then after 90 days the flexural strength of SCGC attains 2.4 MPa, 2.3 MPa, 2 MPa, 1.8 MPa and 1.7 MPa when fly ash (FA) is replaced with GBFS by the weight of 0%, 13%, 28%, 42% and 56% respectively [58].

Furthermore, Fig. 17(b) shows the analysis of the effect of FA content on the tensile strength development of SCGC in which for 1 day the TS of SCGC attains 2.9 MPa, 2.7 MPa, 2.4 MPa and 2.2 MPa when (FA) is replaced with GBFS by weights of 13%, 28%, 42%, and 56%, respectively. Then after 3 days, the tensile strength of SCGC attains 4.7 MPa, 4.5 MPa, 4.2 MPa and 3.5 MPa when FA is replaced with GBFS by the weight of 13%, 28%, 42% and 56% respectively. Then after 7 days, the tensile strength of SCGC attains 5.3 MPa, 5 MPa, 4.8 MPa and 4.3 MPa when FA is replaced with GBFS by the weight of 13%, 28%, 42% and 56% respectively. Also, after 28 days the tensile strength of SCGC attains 5.9 MPa, 5.5 MPa, 5.3 MPa and 5.1 MPa when FA is replaced with GBFS by the weight of 13%, 28%, 42% and 56% respectively. Further after 56 days the tensile strength of SCGC attains 6 MPa, 5.5 MPa 5.3 MPa and 5.1 MPa when FA is replaced with GBFS by the weight of 13%, 28%, 42% and 56% respectively. After 90 days, the tensile strengths of SCGC attains 6 MPa, 5.5 MPa 5.3 MPa and 5.1 MPa when fly ash (FA) is replaced with GBFS by weights of 13%, 28%, 42%, and 56%, respectively. Moreover, Fig. 17(c) shows the analysis of the effect of FA content on the compressive strength development of SCGC in which for 1 day the compressive strength of SCGC attains 30 MPa, 28 MPa, 24 MPa, 23 MPa and 20 MPa when FA is replaced with GBFS by weights of 0%, 13%, 28%, 42%, and 56%, respectively. After three days, the compressive strengths of SCGC attains 51 MPa, 49 MPa, 47 MPa and 40 MPa when FA was replaced with GBFS at weights of 13%, 28%, 42%, and 56%, respectively. After seven days, the compressive strengths of SCGC attains 62 MPa, 58 MPa, 55 MPa, 47 MPa and 34 MPa when FA was replaced with GBFS by weights of 13%, 28%, 42%, 56%, and 70%, respectively. After 28 days, the compressive strengths of SCGC were 68 MPa, 65 MPa, 63 MPa and 55 MPa when FA was replaced with GBFS by weights of 13%, 28%, 42%, and 56%, respectively. After 56 days, the CS of SCGC attains 70 MPa, 68 MPa, 66 MPa, 58 MPa and 53 MPa when FA was replaced with GBFS by weights of 13%, 28%, 42%, 56%, and 70%, respectively. After 90 days, the compressive strengths of SCGC attains 70 MPa, 68 MPa, 66 MPa and 58 MPa when FA was replaced with GBFS by weights of 13%, 28%, 42%, and 56%, respectively [59].

4.1 Microstructures analysis

The synergy between the solution-to-binder (S/B) ratio and micro-silica incorporation is a central determinant of the microstructural maturity and phase evolution of alkali-activated matrices. Figures 18 and 19 offer a direct visual comparison of the two systems: Sample A, designed with a binder content of 500 kg/m³ and S/B = 0.6, and Sample B, prepared with 600 kg/m³ binder and a reduced S/B = 0.5. Both samples contained 15% micro silica, a high-reactivity pozzolanic additive known to modulate nucleation kinetics and refine the pore structure. Despite the identical additive content, the stark morphological contrast between the two samples highlights the non-linear, interdependent effect

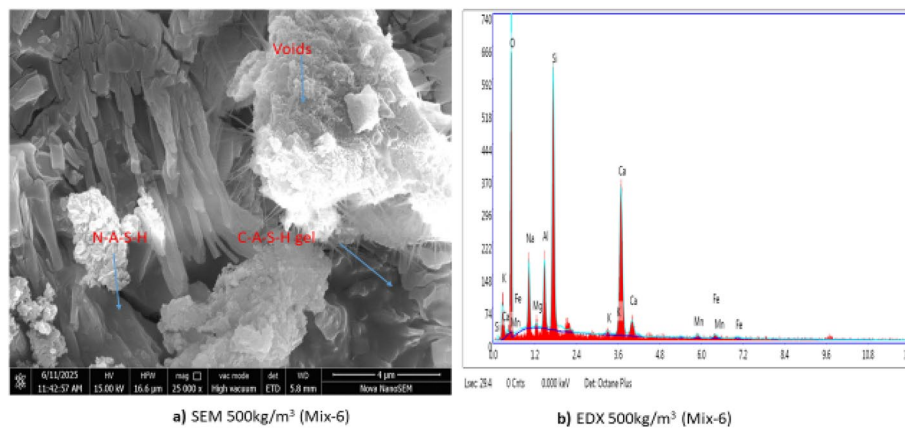


Fig. 18 SEM and EDX image 500 kg/m³ (Mix-6)

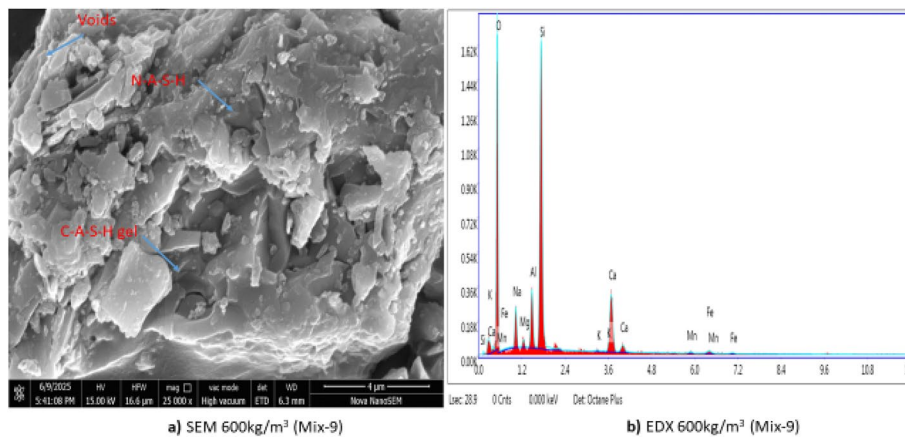


Fig. 19 SEM and EDX image 600 kg/m³ (Mix-9)

of activator saturation and solid-phase availability on the gel connectivity, porosity, and phase homogeneity.

The microstructural evolution of the developed SCGC mixtures was investigated through an integrated SEM, EDX, XRD, and FTIR framework, enabling an epistemically grounded interpretation of the reaction pathways and gel assemblage formation. This multi-evidential approach is crucial because alkali-activated systems are governed by complex dissolution–precipitation mechanisms, in which subtle variations in activator chemistry, precursor reactivity, and solid–liquid ratios govern the topology, chemistry, and continuity of the binding gel network. In the present study, these factors manifested in markedly different microstructural regimes for Mix-6 and Mix-9.

The SEM micrograph of Mix-6 (Fig. 18a) reveals a reaction environment constrained by insufficient polymerization kinetics. The matrix is controlled by angular, incompletely dissolved fly ash and slag residues embedded within a discontinuous gel phase, punctuated by high-contrast voids and poorly connected lamellar reaction domains. Such morphological inhomogeneity is emblematic of systems in which activator oversaturation dilutes the effective ionic concentration, thereby suppressing the dissolution of aluminosilicate glasses and delaying the nucleation of stable oligomer species [51].

This observation was fully supported by the EDX spectrum of Mix-6 (Fig. 18b), which shows a chemical regime rich in Si and Al, but deficient in Ca, accompanied by significant Na incorporation. The resulting high Si/Al and Na/Al ratios and low Ca/Si ratios strongly indicate that the reaction products predominantly comprise N-A-S-H gels, whose polymeric network is governed by charge-balanced Si–O–Al linkages [54]. Such gels, while structurally viable, are known to form frameworks that are more open and typically less prone to nanoscale densification, a feature consistent with the porous and weakly integrated matrix observed in the SEM image.

Mix-9 (Fig. 19a) presents fundamentally different microstructural architectures. Here, the higher binder content and lower solution-to-binder (S/B) ratio fundamentally alter the thermodynamics and kinetics of the alkali activation process. The SEM image revealed a densely packed gel matrix with coherent encapsulation of residual particles and drastically reduced microvoid connectivity. The absence of lamellar debris and presence of tightly fused reaction zones suggest efficient precursor dissolution, rapid gel nucleation, and advanced polymerization. These morphological signatures reflect a system in which the balance between activator molarity and reactive solid content is sufficiently optimized to promote sustained growth of oligomeric species into a dense, three-dimensional aluminosilicate framework [53].

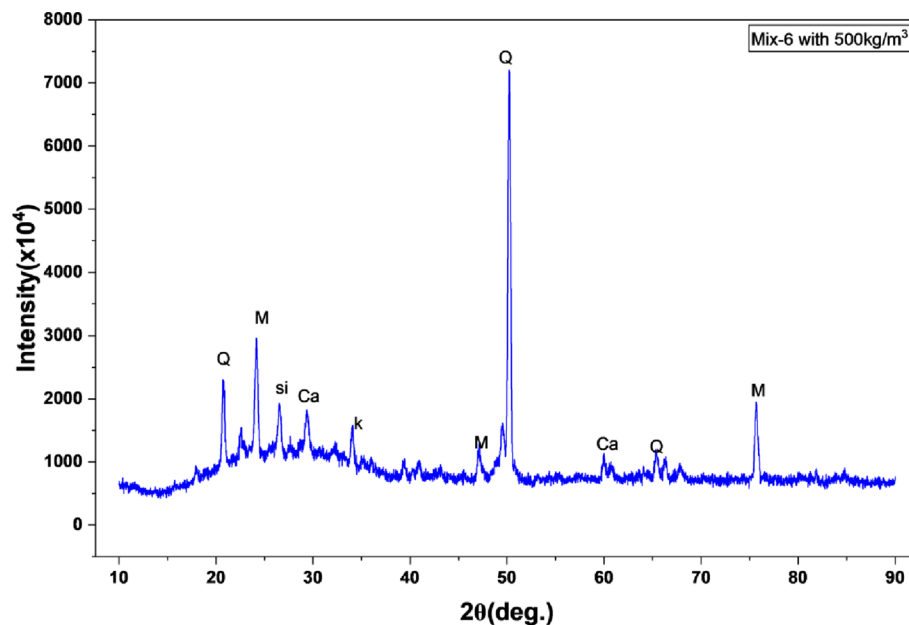
This interpretation was substantiated by the EDX spectrum of Mix-9 (Fig. 19b), which shows a marked elevation in Ca intensities along with strong Si and Al peaks, and moderate Na. Such chemical patterns indicate the emergence of C-A-S-H domains coexisting with N-A-S-H, producing a hybrid gel network. The increased Ca/Si and moderated Si/Al ratios are fully consistent with the partial transformation of sodium aluminosilicate networks into Ca-modified cross-linked gel motifs [54]. The coexistence of these gels is known to enhance structural compactness through dual-phase densification mechanisms, wherein the Na-rich framework provides extensive polymerization, whereas Ca-driven gel phases refine the pore structure and reinforce interparticle bridging.

The C–A–S–H and N–A–S–H labels shown in the SEM images represent interpretations based on combined SEM–EDX–XRD–FTIR evidence, rather than direct SEM phase identification. SEM reveals morphology only; therefore, gel attribution relies on EDX elemental signatures (Ca-rich domains indicating C–A–S–H; Na–Al–Si domains indicating N–A–S–H), supported by XRD and FTIR analyses. The dense gel matrix, reduced voids, and coherent microstructure observed in Fig. 19 are consistent with enhanced polymerization and calcium-assisted gel development, as supported by the SEM morphology, EDX elemental ratios, XRD amorphous halo intensification, and FTIR Si–O–T band shifts. These combined microstructural indicators confirm a more mature hybrid C–A–S–H/N–A–S–H gel network in Mix-9 compared to Mix-6.

It is important to clarify that the contribution of RHA to secondary gel formation cannot be visually isolated or ‘spotted’ in SEM images, because SEM reveals only surface morphology and does not distinguish silica sources. RHA-derived reactive silica participates indirectly in enhancing polymerization, and its influence is therefore inferred from multi-technique evidence; (i) increased Si availability detected by EDX, (ii) a stronger amorphous halo in XRD, indicative of greater gel formation, and (iii) FTIR shifts of the Si–O–T band, reflecting higher polymerization. Thus, while secondary silica gels associated with RHA cannot be visually highlighted, the combined SEM–EDX–XRD–FTIR

Table 5 Representative EDX atomic ratios

Mix	Ca/Si	Si/Al	Na/Al
Mix-6 (500 kg/m ³ , S/B 0.6)	0.75	2.1	0.45
Mix-9 (600 kg/m ³ , S/B 0.5)	1.15	2.8	0.38

**Fig. 20** XRD of image 500 kg/m³ (Mix-6)

signatures, particularly in the more densely polymerized Mix-9 matrix, scientifically support their presence [31].

To support the qualitative microstructural observations, representative EDX atomic ratios were extracted from the selected gel-rich regions of Mix-6 and Mix-9 (Table 5). Mix-6 exhibited a relatively low Ca/Si ratio and moderate Si/Al ratio, indicating a predominantly aluminosilicate-rich gel network. In contrast, Mix-9 showed a higher Ca/Si ratio, together with an increased Si/Al ratio, suggesting enhanced calcium incorporation within a more polymerized aluminosilicate framework. The Na/Al ratios for both mixes remained below unity, indicating a balanced charge compensation within the geopolymer network. These compositional trends collectively suggest the formation of a hybrid C–A–S–H / N–A–S–H- type gel system, consistent with the widely reported alkali-activated binder mechanisms. It is emphasized that these interpretations are inferential and based on combined SEM–EDX–XRD–FTIR evidence, rather than direct phase identification.

4.2 X-ray diffraction (XRD)

XRD analysis provided further structural validation of the distinct reaction environments established in Mix-6 (Fig. 20) and Mix-9 (Fig. 21). Mix-6 exhibits a shallow and weakly defined amorphous hump in the characteristic 25–32° 2θ region, interrupted by pronounced reflections corresponding to quartz, mullite, and intact slag phases. The persistence of these crystalline remnants confirms that the dissolution of the aluminosilicate precursors remained kinetically constrained, consistent with the activator-dilute chemistry suggested by SEM and EDX. The limited broadening of the amorphous

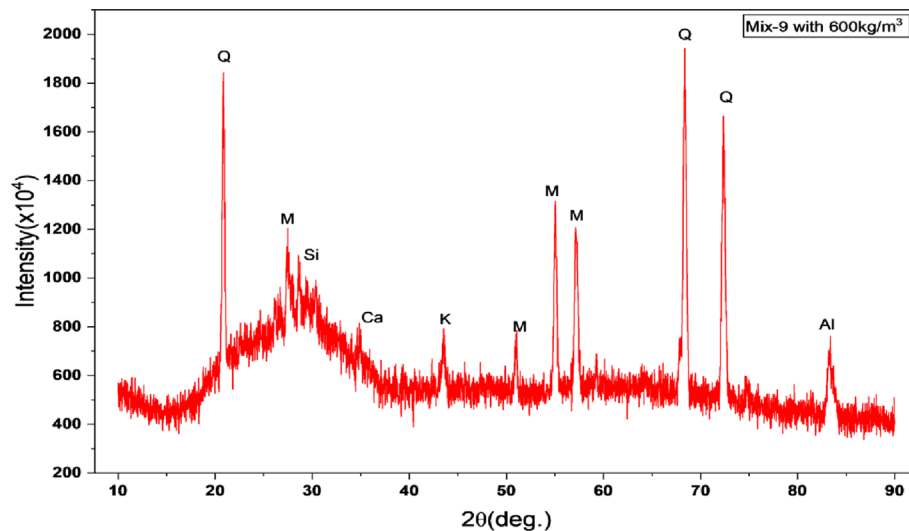


Fig. 21 XRD of image 600 kg/m³ (Mix-9)

envelope signifies the formation of only a modest quantity of reaction gel, indicative of an early-stage geopolymer framework in which polymerization has not progressed sufficiently to mask the crystalline signatures [60].

In contrast, Mix-9 displays a substantially intensified and broadened amorphous halo, a hallmark of advanced geopolymerization, wherein sustained depolymerization of the glassy components is accompanied by extensive precipitation of disordered aluminosilicate and calcium-modified networks [55]. The notable attenuation of crystalline reflections in Mix-9 reflects the more complete breakdown of the precursor phases and the concurrent development of a cohesive, chemically evolved binding matrix. Minor calcite reflections in both mixes arise from atmospheric carbonation, a ubiquitous phenomenon in ambient-cured alkali-activated systems; that does not interfere with the interpretation of reaction progression [56].

4.3 Fourier transform infrared spectroscopy

FTIR spectroscopy provides complementary insights into the molecular configurations of reaction products. In Mix-6, the principal asymmetric Si–O–T (T = Si or Al) stretching band appears near $\sim 955\text{--}960\text{ cm}^{-1}$, indicative of a lightly polymerized aluminosilicate structure in which the degree of Al substitution into the silicate network remains limited. Such vibrational features are characteristic of systems dominated by Na-based charge balancing, where the formation of three-dimensional networks proceeds slowly [61].

In Mix-9, this band underwent a pronounced red shift to $\sim 920\text{--}930\text{ cm}^{-1}$, a well-recognized spectral signature of increased Al incorporation and enhanced network cross-linking, reflecting a more advanced state of polymerization [57]. Additional bands in the $450\text{--}800\text{ cm}^{-1}$ range, corresponding to Si–O–Si and Al–O–Si bending modes, appear sharper and more defined in Mix-9, indicating improved tetrahedral connectivity and the development of a more coherent gel framework. These spectral transformations align closely with the Ca-enriched elemental chemistry detected by EDX and the heightened amorphous content revealed by XRD, confirming that Mix-9 fosters a mature,

chemically heterogeneous reaction matrix in which the N-A-S-H and C-A-S-H domains coexist and interact synergistically [62] (Fig. 22).

Collectively, the SEM, EDX, XRD, and FTIR analyses converge to delineate two fundamentally different mechanistic regimes governing microstructural development. Mix-6 evolved predominantly through Na-mediated aluminosilicate polymerization, generating a loosely packed N-A-S-H matrix characterized by limited gel continuity, restricted amorphization, and a high prevalence of unreacted precursor phases. In contrast, Mix-9 follows an intensified reaction trajectory involving deeper dissolution of the reactive glassy components, extensive formation of amorphous gel phases, and emergence of a chemically validated hybrid C-A-S-H/N-A-S-H network [63]. This hybrid assemblage enhances nanoscale packing, reinforces gel cohesion, and suppresses porosity, culminating in a microstructure that is substantially denser, more chemically integrated, and more mechanically superior (Fig. 23).

5 Conclusion

This study demonstrates that the performance of ambient-cured Self-compacting geopolymer concrete (SCGC) is governed by the combined influence of binder chemistry, solution-to-binder ratio, and coupled reactivity of micro-silica and rice husk ash within a neutral-grade sodium silicate activation system ($M_s = 2.92$). The optimized mixture containing 600 kg/m^3 binder at an S/B ratio of 0.50, delivered superior rheological behavior and significantly enhanced early-age and 28-day compressive strengths without thermal curing. All mechanical properties were obtained from $n \geq 3$ replicate specimens and reported as *mean $\pm$ standard deviation*, ensuring statistical transparency. The strength developments, particularly $\sim 58\%$ and $\sim 85\%$ of the 28-day strength achieved at 3 and 7 days, clearly exceeded the experimental variability, confirming that the observed improvements were statistically meaningful.

Multi-technique microstructural characterization (SEM, EDX, XRD, and FTIR) provided a unified mechanistic explanation for these macroscopic trends. SEM revealed the formation of a densely packed, continuous, and cohesive reaction matrix in the

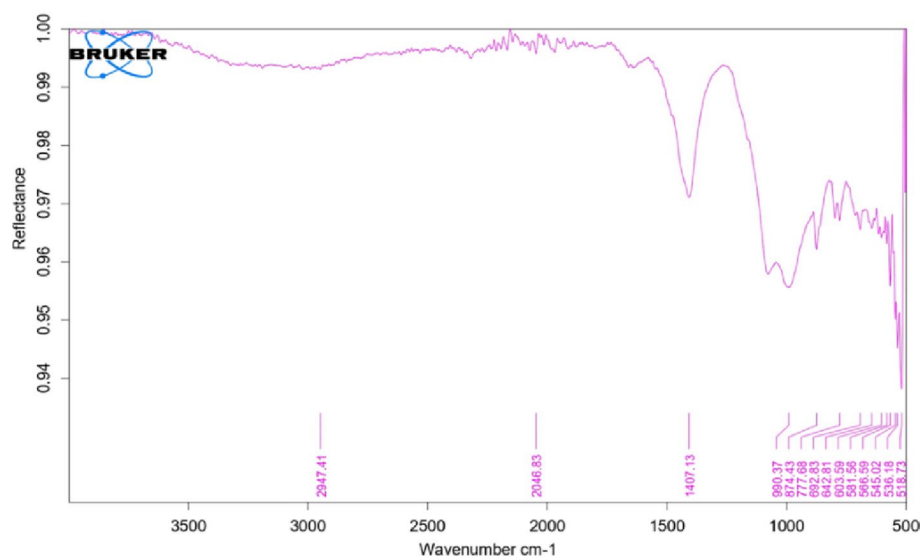


Fig. 22 FTIR of image 500 kg/m^3 (Mix-6)

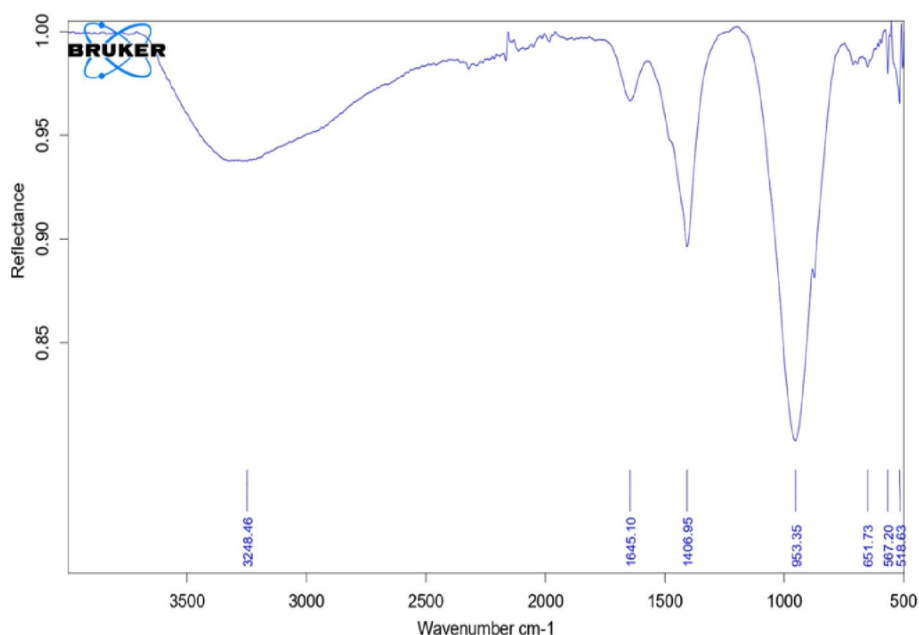


Fig. 23 FTIR of image 600 kg/m³ (Mix-9)

optimized mix, in contrast to the porous and weakly polymerized structure observed in the 500 kg/m³ system. The EDX elemental ratios confirmed the development of a chemically hybrid reaction environment consisting of both Na-aluminosilicate and Ca-modified aluminosilicate gel domains. The XRD patterns exhibited a broadened and intensified amorphous halo, indicating deeper precursor dissolution and extensive amorphous gel formation. The FTIR spectra showed a clear red shift in the Si–O–T asymmetric stretching band, reflecting the increased aluminosilicate network polymerization and enhanced structural refinement. Together, these results verified the formation of a structurally integrated hybrid C–A–S–H/N–A–S–H gel assemblage that fundamentally governs the mechanical superiority of the optimized SCGC.

Overall, the findings confirm that achieving thermodynamically balanced solid–liquid chemistry is critical for producing high-performance self-compacting geopolymer concrete under ambient curing conditions. The convergence of statistically validated mechanical evidence and rigorous multi-scale microstructural analysis provides a strong scientific foundation for the development of next-generation sustainable geopolymer binders.

While the optimized mixture exhibited strong early age and 28-day mechanical performance, the present study did not include durability evaluation, long-term deformation assessment, or structural member verification. Therefore, claims regarding the low-carbon advantages, long-term durability, and suitability for structural and prefabrication applications cannot be asserted at this stage. These aspects lie outside the scope of the present work and require dedicated life cycle, durability, and structural performance investigations in future studies.

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Author contributions

Author Contributions: Conceptualization, Thatikonda Naresh and Mainak Mallik; Investigation, Thatikonda Naresh and Mainak Mallik; Data curation, Thatikonda Naresh, Rajesh Kumar Dora, and Md. Sameer; Formal analysis, Mainak Mallik, S. Venkateswara Rao; Resources, V. V. Praveen Kumar and N. Satya Shiva Prasad; Writing—original draft preparation,

Thatikonda Naresh; Writing—review and editing, Mainak Mallik, S. Venkateswara Rao, and Shiva Prasad; V. V. Praveen Kumar Supervision, Mainak Mallik and S. Venkateswara Rao. All authors have read and agreed to the published version of the manuscript.

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Data availability

All relevant data supporting the findings of this research are included in the manuscript; no additional data are required.

Declarations

Ethics approval and consent to participate

Not applicable, as the study did not involve human or animal participants.

Consent for publication

All authors reviewed the final version of the manuscript and agreed to its submission and publication.

Competing interests

The authors declare no competing interests.

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